

2.7.28

EE25BTECH11016 - Taraka Abhinav

Question:

Find the area of the triangle whose vertices are A(2,3), B(3,5), and C(4,4).

Solution:

Point	Vector
A	$\begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix}$
B	$\begin{pmatrix} 3 \\ 5 \\ 0 \end{pmatrix}$
C	$\begin{pmatrix} 4 \\ 4 \\ 0 \end{pmatrix}$

TABLE 0: Vectors

$$(\mathbf{A} - \mathbf{B}) = \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} - \begin{pmatrix} 3 \\ 5 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \\ 0 \end{pmatrix}, \quad (1)$$

$$(\mathbf{A} - \mathbf{C}) = \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} - \begin{pmatrix} 4 \\ 4 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ -1 \\ 0 \end{pmatrix}. \quad (2)$$

Using (1) and (2), the magnitude of the cross product is

$$\begin{aligned} \|(\mathbf{A} - \mathbf{B}) \times (\mathbf{A} - \mathbf{C})\| &= \sqrt{0^2 + 0^2 + ((-1)(-1) - (-2)(-2))^2} \\ &= \sqrt{0^2 + 0^2 + (-3)^2} = 3. \end{aligned} \quad (3)$$

Therefore the area of triangle ABC is

$$\text{ar}(\triangle ABC) = \frac{1}{2} \|(\mathbf{A} - \mathbf{B}) \times (\mathbf{A} - \mathbf{C})\| = \frac{1}{2} \times 3 = 1.5. \quad (4)$$

Therefore, $\text{ar}(\triangle ABC) = 1.5$.

